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Algorithms as partners in crime: A lesson in ethics by design

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ABSTRACT

The human in the loop is often advocated as a panacea against concerns about AI-powered machines, which increasingly take decisions of consequence in all realms of life. However, can we rely on humans to prevent unethical decisions by machines? We run online experiments modeling both the case where the machine serves as a corrective to the human and where the human serves as a corrective to the machine. Our results suggest that, in the former case, humans make similar decisions whether the corrective is a machine or another human. In the latter case, humans take advantage of rather than correct bad decisions by machines, turning into partners in crime. These findings caution us not to count too much on the human in the loop as a moral corrective. Instead, they tend to argue for human–machine decision-making where the human makes the decision and the machine is the corrective.

1. Introduction

AI-powered algorithms are conquering all realms of life, ranging from mobility to personnel recruitment and penal sentencing (Rahwan et al., 2019). While many feel uncomfortable with leaving consequential decisions to algorithms, hybrid systems, which partner humans with algorithms for decision-making, are more easily accepted (Waldman & Martin, 2022). Hence, the human is attributed a pivotal moral role in this partnership. However, prior research has focused on the morality of the machine rather than the human partnering with it. Thus, there is an active debate about whether algorithms qualify as moral agents (Nyholm, 2018; Purves et al., 2015; Véliz, 2021).[1] In contrast, we know little about how a human's interaction with the machine shapes that human's morality and thus her capability to serve as a moral agent. While some have proposed that machines can improve human morality (Lara & Deckers, 2020), others are concerned that they rather undermine it (Danaher, 2019; Köbis et al., 2021; Vallor, 2015).

The human and the machine are rarely equals in hybrid decision-making. In some cases, the algorithm supports the human, who makes the decision. In other cases, the algorithm makes the decision, and the human is kept in the loop only to intervene if necessary (Rahwan, 2018). The mobility domain illustrates both cases (Danaher, 2019). On the one hand, smart vehicles are equipped with AI-powered advanced driver assistance systems (ADAS) to prevent and correct driving errors and thus to support the driver (e.g., face detection software to measure the driver's fatigue). On the other hand, there are also vehicles about to enter the market which automate the decision-making process and reduce the role of the driver to that of a corrective—in fact, the human in the loop. We examine how the interaction with the algorithm affects the human's behavior depending on whether the human has the role of the decision-maker or the corrective, with the algorithm taking the complementary role of the corrective or the decision-maker.

Whether the human is the decision-maker or the corrective, she has responsibility for the resulting decision. Concerns have been voiced, though, that the very interaction with machines turns humans from moral agents into moral patients, who are unable or unwilling to take responsibility for their decisions (Danaher, 2019). First, partnered decision-making gives the human the opportunity to deflect the responsibility for a morally questionable decision to the algorithm (Köbis et al., 2021). However, while responsibility deflection has been found among humans, the evidence for hybrid partnerships is yet limited (Dana et al., 2007; Kirchkamp & Strobel, 2019). Second, behavioral adaptation, as another mechanism resulting in irresponsible behavior, appears to generalize to hybrid partnerships. Revisiting the mobility example, drivers have been found to compensate the risk reduced by ADAS by taking more risk (Metz et al., 2021). In summary, it is fair to assume that the interaction with the algorithm influences the human's moral behavior.

We propose three research questions about how the partnership with a machine shapes human morality in decision-making. First, we

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[1] We refer to the algorithm as acting, and the human as interacting with it, in a practical sense. For example, an artificial advisor "acts" by giving advice. We do not claim that the algorithm is an agent, let alone a moral agent.

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ask whether a human makes a different decision depending on whether she partners with an algorithm or another human as her corrective to make that decision—that is, in a hybrid versus a human partnership. Second, we examine whether the human's choices differ depending on whether she serves as the decision-maker or the corrective in a hybrid partnership. If she serves as the corrective, the most relevant case is, incidentally, where the algorithm fails and thus correction is most warranted. Third, we examine whether the human's decision differs depending on whether there is a corrective. From a moral viewpoint, there is no reason why human decisions should differ between these three cases. Based on prior empirical evidence, however, they might differ nonetheless. Since this evidence is yet sparse and sometimes inconclusive, we pose research questions rather than make directional predictions.

We conducted three experiments to answer these research questions. In the first experiment, the focal player decides in a power-to-take game (Bosman & van Winden, 2002) how much of a given amount to take from another player. A third player can revise the amount taken downward and thus serves as a corrective. To investigate how human decision-makers respond to artificial versus human correctives, we replace that third player with an algorithm. In the second experiment, the focal player decides how much to take conditionally on the maximum amount that the algorithm allows, with a special interest in the case where the algorithm allows that focal player to take as much as she wants. We thus swap the roles and sequence of events, turning the decision-maker into the corrective (i.e., the human in the loop). In the third experiment, the first two players are left alone to play the basic power-to-take game. Thus, they replicate the interesting case from the second experiment, except that there is no corrective rather than a bad algorithm.

In these experiments, moral behavior is measured in terms of the focal player's request and thus her selfishness (or fairness). The game captures the fact that immoral decisions often involve some personal benefit to the decision-maker at the expense of those affected by the decision. We do not (need to) impose any standard on what would be a fair request (although a request that results in an equal split is certainly a candidate). Instead, we compare the focal players' requests between the three cases. From an ethical viewpoint, again, there is no reason why the focal player should make a different request depending on, first, whether that player's request is potentially corrected by an algorithm or by another human player; second, whether that player is the decision-maker, whose initial request is potentially corrected by an algorithm, or the corrective, who responds to the algorithm's decision to allow her to request and take as much as she wants; third, whether there is an algorithm to correct that player's request to begin with.

Our results suggest, first, that humans respond similarly to an artificial and a human corrective. In our experiments, the amounts requested by our subjects and their expectations about the amounts requested by others were about the same in both cases. Second, if humans are supposed to correct algorithms, they take advantage of the algorithms' bad decisions rather than correct them. Specifically, our subjects requested more in the case where the algorithm allows the human to request and take as much as she wants than in the case where the human first requests an amount, which the algorithm may then correct downward. Third, our subjects requested also more in the case where the algorithm was there but made a bad decision than in the case where there was no algorithm to begin with. This is evidence for undesirable behavioral adaptation, and it suggests that human morality may be worse with a bad machine than without any machine. Our findings cast doubt on whether humans in the loop perform their corrective task as is usually expected.

Our findings hold important lessons for ethically aligned design. They address the question of how to partner algorithms with humans so that they make ethical decisions together. On the one hand, our findings caution us against augmented systems where a largely autonomous algorithm is assisted by a human, who intervenes if the algorithm happens to make an unethical decision. In the very case where human intervention is most needed, the human partner turns into a partner in crime rather than a reliable corrective of the algorithm. We therefore cannot bank on a human in the loop to add value to ethical decisions made by an algorithm. On the other hand, our findings argue for augmented systems where the human plays the active part and makes the decisions, while the algorithm serves as a corrective. Regardless of the quality of the algorithm, humans who are thus partnered with it make as ethical decisions as if partnered with another human. Hence, augmented systems can be designed to benefit from human involvement.

2. Procedure

We ran our three experiments online at a major European university. The laboratory for experimental research in economics of that university maintains a list of volunteers who are regularly invited to participate in economic experiments through the recruitment system ORSEE (Greiner, 2015). These experiments require subjects to perform tasks on networked computers and interact, anonymously, through these computers. Participation is voluntary and paid for. For each experimental session, we arranged a virtual meeting to check our subjects' identities and brief them for the experiment. Together with the ORSEE recruiting system, the identity check precluded that the same subject would participate more than once in one or several of our experiments. We used the oTree software to implement the experiments (Chen et al., 2016). Informed consent was obtained from all subjects, and the experiment was approved by the Institutional Review Board of the German Association for Experimental Economic Research (https://www.gfew.de).

A total of 311 subjects participated in our three experiments. These were randomly assigned to the different roles (e.g., as the decision-maker or the corrective) at the beginning of their experimental session. Throughout their session, the subjects interacted anonymously through the internet, and their compensation was sent to them after their session. The instructions, which are reprinted in the appendix, and all other information needed for the experiment was displayed to them on their computer screens. In case any questions came up (which they did not), the subjects could privately chat with the experimenter during the experiment through the software application that we used to arrange the virtual meeting. In turn, they could not chat with one another and they were muted in that software application to preclude communication between them. We thus prevented our subjects from finding out about other subjects' role, and especially about whom they had interacted with. These settings helped us protect our subjects' anonymity.

Our experiment follows the paradigm of experimental economics that subjects' decisions affect their compensation and thus have monetary implications for them. Thus, the provision of effort or socially desirable decisions are not for free to the subjects in the experiment, but they come at a cost for them, as is the case in real life (Smith, 1994). Experimental economics also holds the norm that subjects must not be deceived by the experimenter, which is enforced by the laboratory where we conducted our experiments. This rule preserves subjects from making decisions that are based on wrong assumptions, especially about their monetary incentives, which experimenters cannot know and which make the results of experiments hard to interpret (Ortmann, 2019). Moreover, we followed the convention of employing a decontextualized setting with a generic decision problem. Neutral framing makes the task easy to understand for subjects and prevents them from slipping in real-life associations (van Dijk & De Dreu, 2021).

The experimental method in general and our experiments in particular have limitations. Thus, our decontextualized decision problem does not require any professional expertise from the human decision-maker or corrective, which enables us, one the one hand, to examine the general causal effect of the design of a human–machine partnership for

decision-making on human morality. On the other hand, however, our experiment does not speak to the role of contextual factors such as the nature or importance of the decision to be made or the decision-maker's level of expertise. Moreover, our subjects (inter)acted anonymously and were not held accountable for their actions. While we have no reason to assume that any of these factors changes our results qualitatively, they might moderate the effects overall. For example, subjects might behave more responsibly in all conditions if they were held accountable. Incidentally, we administered our experiment to a European sample, and it would be interesting to recruit non-European samples.

3. Study 1

Important decisions are often made by teams rather than individuals. The decision-making partners rarely contribute equally to the decision. The decision rules range from judge–advisor systems, where one partner eventually decides after hearing the other's advice, to unanimous consensus (Bonaccio & Dalal, 2006; Humphrey et al., 2002). The dyadic partnership where the first partner decides and the second approves or disapproves and potentially revises that decision falls within this range. In business, for example, large transactions require approval by two managers to come into effect (e.g., the CEO and the CFO). In participative budgeting, the subordinate proposes a budget to the superior, who decides whether to accept, reject, or correct the proposal. In jury trials, the judge decides on the sentence contingent on the jury's verdict. In many cases, one partner makes a decision, while the other serves as a corrective to prevent a bad decision. Hybrid (human–machine) partnerships are a novel instance of partnered decision-making.

How do decision-makers adapt their behavior to such correctives? We can think of two opposite effects. On the one hand, many feel uncomfortable with being criticized or corrected. Decision-makers may therefore be careful to avoid errors and thus correction. By this argument, correctives prevent immoral behavior directly, by correcting decision-makers when they act immorally, and indirectly, by making them act more morally by themselves. On the other hand, decision-makers may become careless or negligent, counting on the corrective to prevent the worst. Responsibility diffusion can explain such behavior (Darley & Latané, 1968; Fischer et al., 2011). The safeguard creates moral wiggle room for decision-makers to deflect responsibility (Dana et al., 2007). If corrected, their immoral actions are inconsequential. Otherwise, they can blame them on the corrective that failed to prevent them. By this argument, correctives still prevent immoral behavior directly, but the indirect effect is that they make decision-makers act less morally by themselves.

Prior research has forcefully argued that machines are not or never can be moral agents, and thus they cannot take moral responsibility (Coeckelbergh, 2020; Constantinescu et al., 2022; Nyholm, 2018; Purves et al., 2015; Véliz, 2021). However, this academic argument does not hinder decision-makers from shifting responsibility to machines and blaming bad decisions on them nonetheless (Feier et al., 2022). Indeed, it is easier to blame a bad decision on a machine than on a human. For example, people feel empathy with other humans more than with machines, and they feel therefore less guilty for wrongly blaming decisions on machines. Moreover, machines do not account for their decisions, and they cannot argue with those who blame them (yet) (Danaher, 2016). Thus, concerns have been raised that partnering with machines undermines moral agency (Danaher, 2019; Vallor, 2015). The limited empirical evidence suggests that decision-makers deflect responsibility to machines and other humans alike (Kirchkamp & Strobel, 2019).

Whether decision-makers deflect responsibility to machines more than humans likely depends on their attitude to machines. There are decision-makers who are suspicious of algorithms (Bigman & Gray, 2018; Dietvorst et al., 2015; Gogoll & Uhl, 2018; Jauernig et al., 2022) and who may want to avoid being corrected by an algorithm and thus confronting it directly. Paradoxically, this leads them to adapt their behavior more readily than to a human corrective. Algorithm-averse decision-makers may also be reluctant to rely on the algorithm as a moral safeguard, deliberate more about what is the right thing to do, and consequently make morally superior decisions. Conversely, decision-makers who appreciate algorithms mind being corrected by a machine less (Logg et al., 2019). They deliberate less because they rely on the algorithm instead. Thus, many drivers engage in riskier and more irresponsible driving behavior when assisted by ADAS (Hiraoka et al., 2010; Metz et al., 2021). In light of these arguments for opposite effects and the limited evidence, rather than make a directional prediction, we pose the following research question.

Research question 1 Suppose that a human partners with either another human or an algorithm to make a decision, where the former human is the decision-maker, and the latter human or the algorithm is the corrective. Does the human decision-maker's moral behavior depend on whether the corrective is human or an algorithm?

3.1. Method

For our experiment, we adapted the power-to-take game. The power-to-take game is similar to the two-player dictator game, where the first player (dictator) is endowed with some amount of money and may then give any part of it to the second player (Forsythe et al., 1994). In the power-to-take game, the second player is the dictator, who may take any share of the initial amount from the first player (Bosman & van Winden, 2002). From an economic viewpoint, both games are equivalent, except that the roles are reversed. Whether the dictator decides how much to give or how much to take does not matter for the resulting distribution of money. From an ethical viewpoint, however, it is considered less acceptable to do harm (take money) than to allow harm to happen (not give money) (Woollard & Howard-Snyder, 2021). The game captures the common trade-off faced by decision-makers, who can serve themselves at the expense of others. This very trade-off naturally raises ethical questions and makes the decision an ethical one.

We endowed the first player (Player A) with 10 ECU (experimental currency units), where 1 ECU converted into €1.20. The second player (i.e., the dictator, Player B) could choose any amount from the set $\{0\ \text{ECU}, 1\ \text{ECU}, \ldots, 10\ \text{ECU}\}$ to take from Player A. Player B's compensation was whatever she took from Player A. Conversely, Player A's compensation was whatever Player B left to him. As Player B's compensation depended on her decision, it was costly for her to leave money to Player A to appear fair. We thus followed the common practice in experimental economics of requiring our subjects to put their money where their mouth was (Falk & Heckman, 2009).[2] The experiment employed two conditions: Human–Human and Human–Algorithm. The Human–Human condition featured another player (Player C), who was paid a fixed compensation of 5 ECU. In addition to their compensation from the power-to-take game, all subjects received a fixed minimum compensation of 2 ECU. The subjects were randomly assigned as Players A, B, or C.[3]

In the Human–Human condition, we introduced the additional Player C, who could cap the amount that Player B could take from Player A. Like Player B, Player C could chose any amount from

[2] The size of the stake is typical of such experiments (Engel, 2011). Variation of the stake size would be one way of manipulating the importance of the decision to the subjects. However, a review of the data from dictator game experiments suggests that the stake size is not a major driver of results (Larney et al., 2019).

[3] The experimental instructions referred to the players as participants. We used a neutral framing so that subjects would engage in ethical decision-making, not in gaming behavior. Again, this is common practice in this kind of experiments (van Dijk & De Dreu, 2021).

the set {0 ECU, 1 ECU, … , 10 ECU}. Player B then received either the amount that she had requested or the amount determined by Player C, whichever was lower. Hence, Player C could correct Player B's decision. Player B could only take the amount that she requested if Player C allowed her to take (at least) that much. We anticipated that Player C would mostly seek an equal distribution and thus cap the amount that Player B could take at 5 ECU. That said, Player C's actual choice is immaterial for us. We are interested in Player B's request, which she makes before learning Player C's decision.

In the Human–Algorithm condition, the role of Player C was assumed by what we described, in the instructions, as an algorithm that imitated human decisions. The algorithm was indeed implemented as a random draw from the distribution of decisions taken by Player-C subjects in earlier sessions of the experiment. Our account of the algorithm imitating human decisions was therefore true. To obtain this distribution, we first ran sessions to administer the Human–Human condition, where we recorded the Player-C subjects' decisions. Subsequently, we ran sessions to implement the Human–Algorithm condition. The decisions of the Player-C subjects in the Human–Human condition were thus available before the sessions of the Human–Algorithm condition were held.

After the subjects had taken their decisions, they were asked about how much they believed other Player-B subjects requested from their Player-A subjects. Note that we asked about the amount that the Player-B subjects requested, not the amount that they obtained, which potentially depended on the Player-C subjects' decisions. The estimates were to be rounded to one decimal. The answer to this question was incentivized with a linear scheme to reward an accurate estimate. For example, an estimate that deviated by 0.1 ECU or less earned the participant 5 ECU; an estimate that deviated by 0.11–0.2, 4 ECU, and so forth. Table 1 lists the payoff for each level of accuracy. The compensation for an accurate estimate came in addition to the fixed compensation of 2 ECU and the compensation from the power-to-take game, which hinged on the decision of Player B and, depending on the condition, Player C or the algorithm.

All players were fully informed about the rules of the game before Players B and, in the Human–Human condition, Player C made their decisions. Player B was never told about Player C's decision. She could infer the exact cap chosen by Player C if she received a lower amount than requested. Otherwise, she could infer that Player C had chosen a cap at or above her requested amount. More importantly, Player C never learned Player B's decision, and Player B knew this. This design choice is necessary to compare the two conditions. In the Human–Human condition, Player B could not engage in image management toward Player C, for example by choosing a lower amount than she desired, because Player C did not learn her decision. In the Human–Algorithm condition, Player B could not reasonably signal anything because the equivalent of Player C, the algorithm, could not read the signal. Incidentally, Player A learned neither Player B's nor, depending on the condition, Player C's or the algorithm's decision.[4]

We are primarily interested in the amount requested by Player B in the Human–Algorithm relative to the Human–Human condition as a measure of the decision-maker's response to the algorithm compared to the human corrective. Player B responds by requesting more or less before she learns the decision of Player C or the algorithm, but knowing that Player C or the algorithm will make a decision on where to cap the amount that she could take from Player A. We also consider Player B's estimate of how much other Player-B subjects requested on average as a measure of her beliefs about others' response. Moreover, we collected some demographic data about our subjects to describe our sample.

[4] Settings where the human corrective learns about the decision-maker's decision are perfectly conceivable. Concerns about her image with the other human might then cause the decision-maker to decide differently (e.g., more morally). We leave the test of this conjecture, which our experiment does not address, for future research.

Table 1
Payoff for estimate.

Deviation	Payoff
>0.5	0 ECU
≤0.5	1 ECU
≤0.4	2 ECU
≤0.3	3 ECU
≤0.2	4 ECU
≤0.1	5 ECU

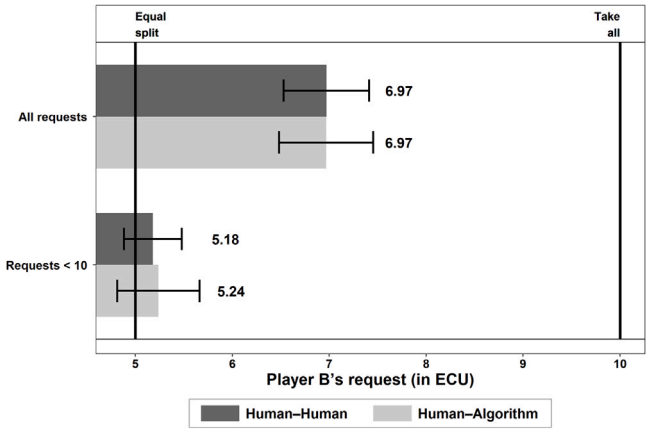


Fig. 1. Mean and standard error of the amount requested by Player B with a human corrective (Human–Human) and an algorithm as corrective (Human–Algorithm).

3.2. Results

A total of 171 subjects volunteered to participate in our first study. Their age averaged 24.9 years (SD 5.14), and 60 percent of them were women. 105 of our subjects were in the Human–Human condition, with 35 playing the role of Player A, B, and C, respectively. The remaining 66 subjects were in the Human–Algorithm condition, one half of them playing the role of Player A; the other half, the role of Player B. Fig. 1 depicts the means and standard errors of the requests of the Player-B subjects. In addition, the requests of less than 10 ECU are reported separately. For completeness, the maximum amount allowed by the Player-C subjects, which is not depicted, averages 5.66, with a standard error of 0.33, and the median and modal maximum amount allowed are both 5 ECU. Although there were Player-C subjects who would not allow Player B to take anything or who would allow her to take the whole amount (i.e., who set a cap of 0 or 10 ECU), these were outliers. The amounts chosen by the Player-C subjects cluster around 5 ECU.

Do decision-makers respond differently to an artificial as opposed to a human corrective? The results of the experiment suggest that they do not. The Player-B subjects' requests are the same, on average, whether there is a human Player C or an algorithm to correct their requests downward (both 6.97, $t = 0.003$, $p = 0.998$). To reveal potential underlying differences, we consider the requests of less than the maximum 10 ECU separately. The maximum 10 ECU was requested by about a third of the Player-B subjects in both the Human–Algorithm and Human–Human conditions (36.4 and by 37.1 percent). The remaining requests come close to an equal split, on average. Again, they do not differ between the conditions ($5.18 < 5.24$, $t = 0.11$, $p = 0.914$). Incidentally, the Player-B subjects' estimates of others' requests do not differ between the conditions either ($6.08 > 5.98$, $t = 0.19$, $p = 0.851$). We conclude that decision-makers respond similarly to an algorithm and to a human corrective, and they believe that others do the same.

Prior evidence about the role of algorithms in the moral domain is mixed. While humans are skeptical about algorithms making ethical decisions (Bigman & Gray, 2018; Gogoll & Uhl, 2018; Jauernig et al., 2022), they readily take moral advice from an algorithm even when

skepticism is warranted (Krügel et al., 2022). Likewise, while decision-makers exploit moral wiggle room to deflect responsibility (Dana et al., 2007), they deflect it to an algorithm or another human alike (Kirchkamp & Strobel, 2019). There is no direct evidence that decision-makers are more uncomfortable with being corrected by another human rather than an algorithm, and the available indirect evidence did not suggest a directional prediction. The result that decision-makers are indifferent, and that they expect the same from others, is therefore not a surprise. Having established this baseline, we turn to the question of whether a human, in a hybrid partnership, acts differently as a corrective to an algorithm rather than as a decision-maker with an algorithm as corrective.

4. Study 2

Partnered decision-making can be organized in different ways. In Study 1, we considered the case where the decision is made by a human and the algorithm serves as a corrective, such as in ADAS. Despite concerns about responsibility diffusion and risk compensation, this use of AI-powered algorithms is mostly welcomed and rather uncontroversial. Moreover, our results suggest that such effects do not differ from those that occur in a partnership where both the decision-maker and the corrective are human. Conversely, people feel less comfortable with the idea that an algorithm decides on important issues alone, such as whether they will receive a loan; how much tax they will pay; whether they will get the job they have applied for, to use but few examples of decisions made by algorithms (Rahwan et al., 2019). Decision-making by algorithms is more acceptable with human participation. Thus, it is commonly claimed that there be a human in the loop for decisions of import, so that such decisions are not made by algorithms alone (Burton et al., 2020).

Revisiting the Human–Algorithm condition of Study 1, where the algorithm serves as a corrective to the human decision, the concept of the human in the loop describes the reverse case of partnered decision-making. The human turns into the corrective to the algorithm's decision. As opposed to AI-powered ADAS, for instance, which support the driver and prevent and correct errors that she might commit, the driver of a fully automated vehicle would not normally tamper with the software running her car. Mostly a passenger more than a driver, she is supposed to intervene only if necessary. These two cases of partnered decision-making—with the human decision-maker and the algorithm as the corrective or the algorithm as the decision-maker and the human as the corrective—differ from both a psychological and a practical viewpoint. It is therefore interesting to contrast them with regard to the question of how the human in the decision-making team discharges her duty. Does she actually correct bad, or unethical, decisions of the algorithm?

Again, the participation of the algorithm in decision-making creates potential for responsibility diffusion and thus moral wiggle room for the human decision-maker (Dana et al., 2007). In Study 1, however, the human moves first and makes the decision; as the corrective, she moves second, if at all. It is easier to attribute the responsibility for a morally questionable outcome to the algorithm in this case. After all, the human decision-maker fails to prevent the outcome rather than causes it (Woollard & Howard-Snyder, 2021). From a psychological viewpoint, the algorithm's decision creates a status quo, which the decision-maker is biased to anchor on (Kahneman et al., 1991; Samuelson & Zeckhauser, 1988). The first-mover role also makes the algorithm salient, and the human decision-maker is led to think carefully about what to make of the algorithm's decision when making her own. It is less straightforward to shift responsibility on to a corrective that possibly intervenes after the fact and that thus fades into the background.

In light of these differences, it is conceivable that the decision of the human differs depending on whether she is supposed to react as a corrective, or whether she is required to act. The reversed sequence of events and the increased salience of the algorithm make it convenient for the human partner to go along with the algorithm's decision rather than correct it. Incidentally, although people feel uncomfortable with an algorithm making an important decision alone, it seems that they care more about having influence on the algorithm's decision than about how much influence they have. Specifically, prior research shows that even minimal leverage on the algorithm's decision is enough to reduce the algorithm aversion carried by the human (Dietvorst et al., 2018). This does not imply that the human in the loop will not use the leverage that she has on the partnered decision, but it certainly does not suggest either that she will. The Human–Algorithm condition of our first study offers a benchmark to answer the following research question.

Research question 2 Suppose that a human partners with an algorithm to make a decision, where either the human is the decision-maker and the algorithm the corrective, or the algorithm is the decision-maker and the human is the corrective (i.e., the human in the loop). Does the human's moral behavior depend on whether that human is the decision-maker or the corrective?

4.1. Method

Our second experiment resembled the Human–Algorithm condition of our first experiment. Again, we had two players and an algorithm interact for partnered decision-making. Player A was endowed with 10 ECU; Player B requested an amount from the set $\{0\ \text{ECU}, 1\ \text{ECU}, \ldots, 10\ \text{ECU}\}$ from Player A; the algorithm capped the amount that Player B could take. Recall that, in the Human–Algorithm condition of our first experiment, Player B and the algorithm decided independently on their amounts. If the algorithm's cap was binding, Player B learned what the cap was after stating her request because she obtained the smaller of the two amounts. If the algorithm's cap was not binding, Player B could only infer that the algorithm had allowed her to take at least as high an amount as the one she had requested. In both cases, Player B did not know the algorithm's decision when she made hers.

In our second experiment, in turn, Player B was asked to request an amount conditional on the cap chosen by the algorithm, for each cap that the algorithm could possibly choose. That is, Player B decided whether to request 0 or 1 ECU from Player A, assuming that the algorithm capped the amount that she could take at 1 ECU; 0, 1, or 2 ECU, assuming that it capped this amount at 2 ECU, and so forth. The algorithm's possible choices resulted in eleven cases, for ten of which Player B was asked to indicate the amount that she would request. (In the case where the cap was 0 ECU, Player B could not do anything but request 0 ECU, which we did not ask her to indicate.) Subsequently, one of these eleven cases was randomly drawn from the distribution of decisions taken by the Player-C subjects in the Human–Human condition of Study 1. Hence, the algorithm could again be said to emulate human behavior, and like in the Human–Algorithm condition of Study 1, it was also introduced as such to the subjects who participated in Study 2.

This procedure, the so-called strategy method (Selten, 1967), changes the sequence of events compared to Study 1. Suppose, for example, that the algorithm decides to cap the amount that Player B may take from Player A at 7 ECU. In Study 1, Player B decides how much to request before she learns that the cap is 7 ECU. (Indeed, she only learns that the cap has been 7 ECU if she requests more than 7 ECU.) In Study 2, Player B knows the cap before she decides how much to request because she makes her decision under the hypothesis that the cap is 7 ECU. Hence, Player B's request follows the algorithm's choice of the cap, and she responds to this choice. Study 2 thereby turns Player B into the corrective to the algorithm, whereas the algorithm is the corrective to Player B in Study 1. We label the only condition of Study 2 the Algorithm–Human condition and contrast it with the Human–Algorithm condition of Study 1. Incidentally, the strategy

method equips us with additional data, as it yields Player B's response in all possible cases.

We are particularly interested in how much Player B requests in the case where the (hypothesized) cap is 10 ECU.[5] We consider this the case of a "bad" machine, which fails to fulfill its function of protecting Player A. Although any moral standard for how much Player B should take can be contested, 5 ECU, which results in an equal split, is a likely candidate. Clustering around 5 ECU, the amounts chosen by the Player-C subjects in Study 1 suggest this as a descriptive norm (Bicchieri, 2006, pp. 29–34). They also argue for our reading that not capping the amount that Player B may take is morally a "bad" decision. Most Player-C subjects consider it their task to intervene and protect Player A. Ethically, there is no reason why the Player-B subjects should request more in case of absence of a binding cap in the Algorithm–Human condition of Study 2 than in the Human–Algorithm condition of Study 1. In both cases, Player B can choose any amount of up to 10 ECU, and the presence or absence of a corrective does not justify a different choice.[6]

Like in Study 1, we elicited the subjects' beliefs about how much the other Player-B subjects requested, in addition to collecting their requests. Since our focal case is the one where the algorithm does not cap the amount that Player B may take from Player A (i.e., where the algorithm opts for a cap of 10 ECU), we asked the subjects to estimate how much other Player-B subjects requested under the hypothesis that the algorithm chose a cap of 10 ECU. We employed the same linear incentive scheme as in Study 1, which is depicted in Table 1, to reward accurate estimates.

4.2. Results

We had 70 subjects who participated in our second study. The subjects' mean age was 24.4 years (SD 3.21), and 57 percent were women. Half of the subjects assumed the role of Player A; half, of Player B. Fig. 2 depicts, on the one hand, the means and standard errors of the requests of the Player-B subjects in the Human–Algorithm condition of Study 1 as a benchmark for comparison. On the other hand, it shows the same statistics for the Player-B subjects' requests in the Algorithm–Human condition of Study 2—in the focal case where the algorithm sets the maximum amount that Player B may take from Player A to 10 ECU. In this case, the algorithm does not constrain Player B indeed. Again, we report the summary statistics for the requests of less than the maximum 10 ECU separately (i.e., the algorithm allows Player B to take 10 ECU, but Player B nonetheless requests less than 10 ECU). We note differences in the means between the two conditions, both when including and when excluding the requests of 10 ECU.

Does the human's behavior depend on whether she is the decision-maker or the corrective? That is, does the human in the loop intervene if the algorithm does not fulfill its function and protect Player A? Again, there is no reason why Player B should request more if she makes her own than if she corrects the algorithm's decision. Nonetheless, the Player-B subjects did request more in the Algorithm–Human than in the Human–Algorithm condition ($6.97 < 8.29$, $t = 2.17$, $p = 0.034$). 54.3 percent requested the maximum 10 ECU in the focal case of the Algorithm–Human condition, as opposed to 36.4 percent in the Human–Algorithm condition, and the requests of those demanding less

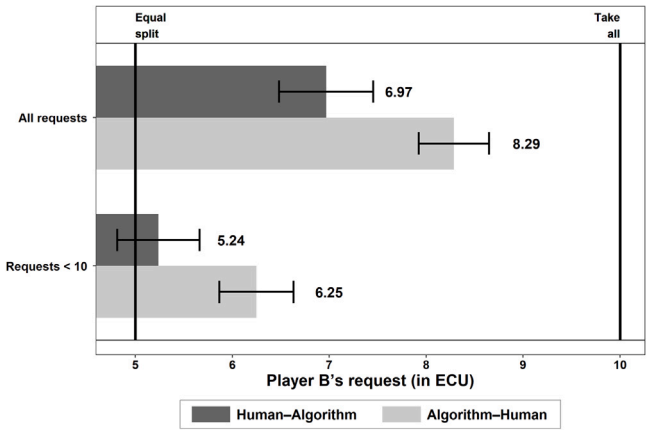


Fig. 2. Mean and standard error of the amount requested by Player B when acting as a decision-maker with an algorithm as corrective (Human–Algorithm) and as a corrective to an algorithm that does not constrain her action space (Algorithm–Human).

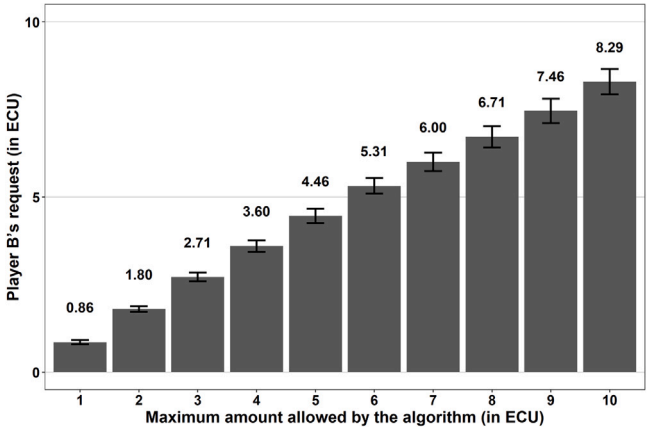


Fig. 3. Mean and standard error of the amount requested by Player B for each possible maximum amount that the algorithm can allow in the Algorithm–Human condition.

were still higher ($5.24 < 6.25$, $t = 2.267$, $p = 0.085$). Hence, the difference is driven by more Player-B subjects requesting more of Player A's money. Incidentally, the Player-B subjects' expectations about others' requests follow this pattern ($5.98 < 8.09$, $t = 4.33$, $p < 0.001$). Summing up, the results suggest that the human in the loop fails in her task.

Fig. 3 depicts the mean request and standard error for each potential cap higher than zero. It turns out that the requests decrease relative to the maximum amount available and the variance increases as the (hypothetical) cap increases. That is, if Player B may request 0 or 1 ECU, under the hypothesis that the algorithm will cap her request at 1 ECU, almost all Player-B subjects request 1 ECU. If Player B may request any amount of up to 10 ECU, the Player-B subjects vary in how much they request, some requesting less than the maximum 10 ECU. Thus, the amount requested decreases relative to the maximum amount allowed, as the latter increases, while the variance increases. That said, even in the case where the maximum amount allowed is 10 ECU, the average request comes close to this amount. Hence, the Player-B subjects show a strong tendency to take whatever the algorithm allows rather than serve as a corrective, which further supports the conclusion that the human in the loop is not a reliable corrective.

In summary, our results cast doubt on whether humans are reliable correctives to decisions by algorithms. While we should expect that they apply the same moral standards as correctives and decision-makers, they act more self-interestedly and thus irresponsibly as correctives according to their own standards. Most of our Player-B subjects

[5] As an alternative to the strategy method, we could present each Player B in Study 2 with one specific cap, where the probability of that cap derives from the distribution of the choices of the Player-C subjects in Study 1. Given our interest in the case of the 10-ECU cap, however, this would be a wasteful procedure.

[6] We might argue that there is nothing "bad" about the algorithm's non-intervention, which gives full decision authority to the human. However, this does not change anything about our finding of (selfish) behavioral adaptation and the implication that the mere presence of an algorithm results in less moral behavior, by the descriptive norm established by the Player-C subjects.

just went along when the algorithm failed to protect Player A and many of the others made rather small corrections. The resulting mean request is significantly higher than in the Human–Algorithm condition, although there is no moral reason why the Player-B subjects should get more in the Algorithm–Human condition. Likewise, the mean request is also higher than allowed by the impartial Player-C subjects, which suggests a descriptive norm of not taking much more than half of Player A's endowment.

5. Study 3

The Human–Algorithm and the Algorithm–Human partnerships from Studies 1 and 2 are both examples of hybrid decision-making, which partners a human with a machine. They differ in whether the decision-maker is human and the corrective is a machine or whether the opposite is the case. From a moral viewpoint, this difference does not warrant that the human requests more in one case than the other. Nonetheless, Study 2 shows that the human requests more as a corrective than as a decision-maker. In the case where correction is most warranted, she exploits the algorithm's failure to constrain her action space rather than corrects this failure. Thus, the comparison turns out unfavorable for the Algorithm–Human partnership. However, there is not a binary choice between the Human–Algorithm partnership, on the one hand, and the Algorithm–Human partnership, on the other. Leaving the decision to the human decision-maker alone is normally a feasible alternative. Incidentally, this alternative is also interesting from a theoretical viewpoint.

While the Algorithm–Human partnership from Study 2 compares unfavorably to the Human–Algorithm partnership from Study 1, we have not benchmarked it against the absence of any corrective yet. Although the average request of 8.29 ECU in the case where the algorithm does not constrain the decision-maker is close to the maximum amount of 10 ECU, we wonder how much a human decision-maker would request (and, in this case, take) if there were no algorithm and thus no constraint. It is conceivable that it is worse to have the human correct the algorithm than to have the algorithm correct the human, but that it is still better than to have no corrective. It goes without saying that any corrective is better than none for Player A, because it corrects a selfish decision of Player B once Player B has made such a decision. More interesting is the indirect effect of the corrective on Player B's morality—i.e., how much money she requests depending on whether there is a corrective or none. Hence, we state the following research question.

Research question 3 Suppose that a human either partners with another human or an algorithm to make a decision, where the former human is the decision-maker and the latter human or the algorithm is the corrective, or the human makes her or his decision without any corrective. Does the human's moral behavior depend on whether there is a corrective (another human or an algorithm)?

5.1. Method

We conducted one more experiment, where we administered the standard version of the power-to-take game (Bosman & van Winden, 2002). Again, Player A was endowed with 10 ECU, and Player B could request any amount from the set $\{0\text{ ECU}, 1\text{ ECU}, \ldots, 10\text{ ECU}\}$ from Player A. This time, there was neither a Player C nor an algorithm to constrain Player B's action space, and thus Player B obtained any amount that she requested from Player A. Consistent with our labeling of the previous three conditions of Studies 1 and 2, we label this only condition of Study 3 the Human condition. Like in the first two studies, we elicited the Player-B subjects' beliefs about other Player-B subjects' requests. Since Player B would always obtain what she requested, we asked the subjects for their estimate of what other Player-B subjects would take (rather than request) from their Player A. Incidentally, the incentive scheme to reward the subjects for making an accurate estimate was the same as in the first two studies, as depicted in Table 1.

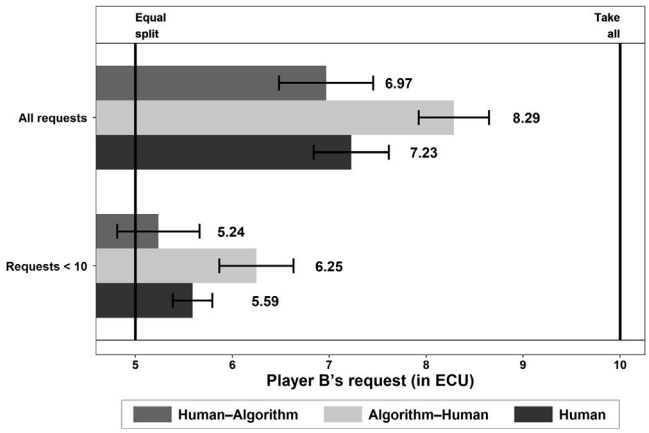


Fig. 4. Mean and standard error of the amount requested by Player B when acting as a decision-maker with an algorithm as corrective (Human–Algorithm); as a corrective to an algorithm that does not constrain her action space (Algorithm–Human); as a decision-maker without any corrective (Human).

The Human condition offers a benchmark for the focal case of the Algorithm–Human condition, where the algorithm does not cap the amount that Player B may take. In both cases, Player B can take as much as she wants. However, the Human condition can also be contrasted with the Human–Algorithm condition of Study 1. On the one hand, decision-makers might request less in the presence of a corrective to avoid correction and because they make more responsible decisions without a safeguard to rely on. (The less they requested, the less likely they were corrected.) On the other hand, they might shift on the responsibility for an excessively high request to the corrective and thus, intentionally or negligently, request even more to serve themselves if there is a corrective (Dana et al., 2007). We compare the amounts requested by the Player-B subjects in the Human–Human and Human–Algorithm conditions (both with corrective) to those in the Human condition (without corrective) to test whether any of those two effects prevails.

5.2. Results

70 subjects participated in our third study. The subjects' age averaged 24.5 years (SD 5.43), and 67 percent were women. Half of the subjects assumed the role of Player A; half, of Player B. Fig. 4 shows the means and standard errors of the Player-B subjects' requests, first, in the Human–Algorithm of Study 1; second, in the case of the Algorithm–Human condition of Study 2 where the algorithm does not constrain the amount that Player B may take; and third, in the Human condition of Study 3. As usual, we report the summary statistics separately for the requests of less than 10 ECU. In the figure, we observe that the requests in the Human–Algorithm and the Human conditions are quite similar, whereas those in the Algorithm–Human condition are higher than those in the other two conditions.

Study 2 shows that Player-B subjects request more from Player A in the Algorithm–Human than the Human–Algorithm team. Do they still request less than in the Human condition, which does not offer any protection to Player A? The answer to this question is negative. The Player-B subjects' requests are higher in the Algorithm–Human than in the Human condition ($8.29 > 7.23$, $t = 1.99$, $p = 0.050$), and so are their estimates of other Player-B subjects' requests ($8.09 > 6.66$, $t = 3.49$, $p < 0.001$). Both the percentage of Player-B subjects requesting 10 ECU and the average of the remaining requests are higher in the focal case of the Algorithm–Human than in the Human condition ($54.3 > 37.1$ percent, $6.25 > 5.59$ ECU). The game played in those two conditions was

technically the same (a dictator game where Player B's action space was unconstrained) except for the presence of an algorithm, which would not intervene on behalf of Player A. We conclude that the presence of the algorithm exacerbates Player B's self-serving behavior.

Returning to the results of Study 1, does Player B request more or less if there is a corrective compared to if there is none? Although Fig. 4 shows that the Player-B subjects' requests are slightly higher on average if there is an algorithm than if there is none, this difference is not statistically significant, neither for all requests taken together (7.23 versus 6.97 ECU) nor for the requests of less than the maximum 10 ECU (5.59 versus 5.24 ECU). This result is mirrored by the Player-B subjects' estimates of the other Player-B subjects' requests, which are insignificantly lower in the Human–Algorithm than in the Human condition as well (5.98 versus 6.66 ECU). Statistically, the Player-B subjects do not request more or less with an algorithm as a corrective than without, and they do not believe that others do so. Hence, we neither find evidence to corroborate the argument that human decision-makers engage in irresponsible behavior if there is a corrective nor for the conjecture that they behave more morally to avoid correction.

These results should not be mistaken to show that the corrective does not help Player A (Human–Algorithm versus Human condition) or that it even hurts him (Algorithm–Human condition). The Player-A subjects keep more of their endowments in all cases of team decision-making. First, their average payoff is higher in the Human–Human than in the Human condition, where there is no corrective ($4.89 > 2.77$, $t = 4.41$, $p < 0.001$). Since the algorithm was modeled on the Player-C subjects' decisions, the same holds true for the Human–Algorithm condition ($5.18 > 2.77$, $t = 4.50$, $p < 0.001$) and also the Algorithm–Human condition, where all eleven cases rather than the focal case only matter for Player A's payoffs ($4.86 > 2.77$, $t = 4.02$, $p < 0.001$). Our results are about the effect of the algorithm on the decision-maker (i.e., behavioral adaptation), not about the effect on the person affected by her decision (Player A). While the algorithm helps Player A, it has an undesirable effect on the human decision-maker (Algorithm–Human).

6. Conclusion

In summary, we conducted three experimental studies to explore how hybrid decision-making, which partners a human with an algorithm, shapes human morality. We distinguished two categories of partnered decision-making, depending on whether the human makes the decision and the algorithm is the corrective or the algorithm makes the decision and the human is the corrective. This difference turns out crucial for the effect of the partnership on human morality. Human decisions are similar, regardless of whether the algorithm, another human, or neither is there as a corrective. However, if the algorithm makes the decision and the human is the corrective (i.e., in the typical case of the human in the loop), we see little human correction when it is most warranted. Indeed, the human takes advantage of the algorithm's decision rather than corrects it. Despite this unfavorable effect of the partnership on human morality, either corrective benefits the third party affected by the decision on average, though, as it corrects some bad decisions.

We contribute to research on ethics by design. While this research has long focused on the morality of the machine (Anderson & Anderson, 2011), the question of how partnering with machines affects human morality is no less relevant (Danaher, 2019; Vallor, 2015). Partnered decision-making calls for research that considers the machine, the human, and how they interact in moral decision-making. Our findings add to the concern that bad machines may corrupt good morals (Köbis et al., 2021). From the viewpoint of virtue or Kantian ethics, which rely on individuals to act morally (i.e., virtuously or dutifully), this insight is alarming. The good news is that hybrid decision-making can be designed to avoid this effect. Humans serve themselves less if they do not know whether their decision will be corrected by an algorithm than if they know that it will not be corrected by it. That is, we do not observe moral corruption if the human is left uncertain about the algorithm's decision and thus decides independently. From the consequentialist viewpoint of utilitarian ethics, it is reassuring to see that ethically aligned design can preserve human morality.

Our findings have practical implications for the design of hybrid partnerships. They first caution against relying on the human in the loop to correct errors induced by an algorithm—that is, partnerships where the algorithm makes the decision and the human serves as a corrective. Instead of correcting a bad decision, the human turns into a partner in crime, who exploits the moral wiggle room left by the algorithm to serve herself. This is not a case against hybrid decision-making, though, but a call to explore ways of enabling the human in the loop to function as a corrective. For example, correcting a decision is costly, and it is thus self-serving to go along. In Study 2, the human was accordingly best served by not correcting the algorithm. The lower the cost of correction, the more likely the human corrective is to function. To curb the human's self-interest, she could also be subject to correction, possibly on a random basis, by another corrective. From the results of Study 1, this second corrective could be human or a machine.

We should note that our experiment is subject to limitations, which suggest avenues for future research. The decontextualized setting allows us, on the one hand, to identify the general causal effect of the hybrid partnership on human decision-making. On the other hand, however, it is conceivable that contextual factors, such as the decision on hand or the level of expertise of the human in mitigate this effect. Likewise, the effect might be moderated by individual differences, such as the attitude to algorithms supporting human decision-making in the moral domain. Moreover, it is likely that accountability, which is absent from our experiment, induces more responsible decisions, whether the human serves as the decision-maker or the corrective, and whether she partners with an algorithm, another human, or neither. Incidentally, while it seems reasonable to run an experiment like ours with European subjects, who have likely been exposed to algorithms, it would be interesting to examine whether the results are the same with a non-European sample.

It is now commonplace that moral behavior is contingent on the context (Bicchieri, 2006). People develop sophisticated strategies to maintain a positive self-image while they make morally questionable decisions, taking advantage of any situational ambiguity available to them (Dana et al., 2007). Prior research shows that people are able to use even simple devices to justify immoral decisions to themselves. For example, in a series of experiments, subjects who flipped coins to assign tasks fiddled with these coins to assign themselves interesting and others dull tasks (Batson et al., 1997, 2002, 1999). Partnering with algorithms for decision-making creates much more and subtler opportunities to hide or justify immoral behavior than coin fiddling ever can (Feier et al., 2022). Our studies show that algorithms can abet immoral behavior. While hybrid partnerships for decision-making have a tremendous potential, we should be aware of the repercussions of partnering with an algorithm on human morality when designing hybrid partnerships.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.chb.2022.107483.

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